

## **Introduction: Validating a Bait-Capture Enrichment Protocol to Analyze Transcriptional Responses to Chemical Contaminants in Zebrafish**

Urban growth and the expansion of industrial practices have resulted in large amounts of synthetic chemicals entering aquatic ecosystems (Piwowarska & Kiedrzyńska, 2022). This threatens the health of many aquatic species, including fish, and has been associated with impaired immunity, reproductive health, and increased mortality (Kime, 1999). To tolerate chemical contaminants, aquatic species rely on specialized proteins that metabolize and eliminate toxic compounds (Eide et al., 2021; Goldstone, 2008; Goldstone et al., 2006). The subset of genes that confer tolerance to toxic elements is called the chemical defensome, and encodes a diverse array of proteins, including transcription factors, receptors, transporters, and antioxidants (Eide et al., 2021; Goldstone, 2008, Goldstone, 2006).

The Wilson lab is interested in how various chemicals affect the development of zebrafish and which pathways zebrafish use to detoxify and export harmful chemicals. By studying zebrafish, the lab hopes to identify important enzyme pathways involved in chemical metabolism, which have applications in metabolic mechanisms that could be used to predict effects of newly developed chemical compounds in aquatic environments. Using a transcriptomic approach, the lab aims to investigate the expression patterns of ~595 known chemical defensome genes in male and female zebrafish, across several tissues, following exposure to chemical contaminants. By performing this analysis, the hope is to observe how the expression of chemical defensome genes changes over time and across tissues, to understand how zebrafish metabolize specific chemicals, and whether specific classes of chemical contaminants pose greater risks in aquatic environments due to inability to metabolize these compounds.

Generating transcriptomes for several tissues, for both male and female zebrafish, exposed to several chemical contaminants, at multiple timepoints, becomes prohibitively expensive. Additionally, we are only interested in ~5% of the total transcripts, and several genes of interest are low-abundance and difficult to detect with conventional RNA-sequencing protocols (Flint et al., 2023). Therefore, the lab has opted to use a bait-capture enrichment protocol. Bait-capture enriches RNA samples for specific gene targets using specially designed probes, allowing for greater resolution of low-abundance transcripts at reduced sequencing depths (Sakkos et al., 2026). However, because this technique required the design of a novel probe set to target the 595 defensome genes, and has not been used in this system before, we needed to validate that this protocol did not distort the representation of transcript abundances.

### **Data**

The data consists of bait-capture enriched and unenriched RNA-sequencing data, generated from the same RNA libraries of male and female zebrafish grown under normal conditions. There are three samples for male zebrafish, and three samples for female zebrafish, each representing distinct biological replicates. Each unenriched sample is paired with an enriched sample, as the same source material was used for both.

### **Question and Objectives**

Does the bait-capture enrichment distort the abundances of transcripts, as compared to conventional RNA-sequencing methods?

1. Compare unenriched and enriched RNA-sequencing counts for the target genes, to ensure proportionality is maintained following bait-capture enrichment

- a. Use an equivalence test (two one-sided t-tests; TOST) on the slope of the regression model to determine equivalence to 1
  - b. Use a minimal effects test (MET) on the intercept of the regression model to demonstrate that the slope is clearly greater than zero.
2. Compare expression patterns of individual genes to identify distorted counts that should be disregarded in subsequent analyses
  - a. Perform paired TOSTs to compare relative expression values of enriched and unenriched data, for every transcript in the target list
  - b. Perform METs to identify true ‘non-equivalent’ transcripts
3. Perform differential gene expression analysis between male and female tissue for enriched and unenriched data, respectively
  - a. Compare DEG calls from enriched and unenriched data
  - b. Determine if bait-capture enrichment provides better resolution for low-abundance transcripts

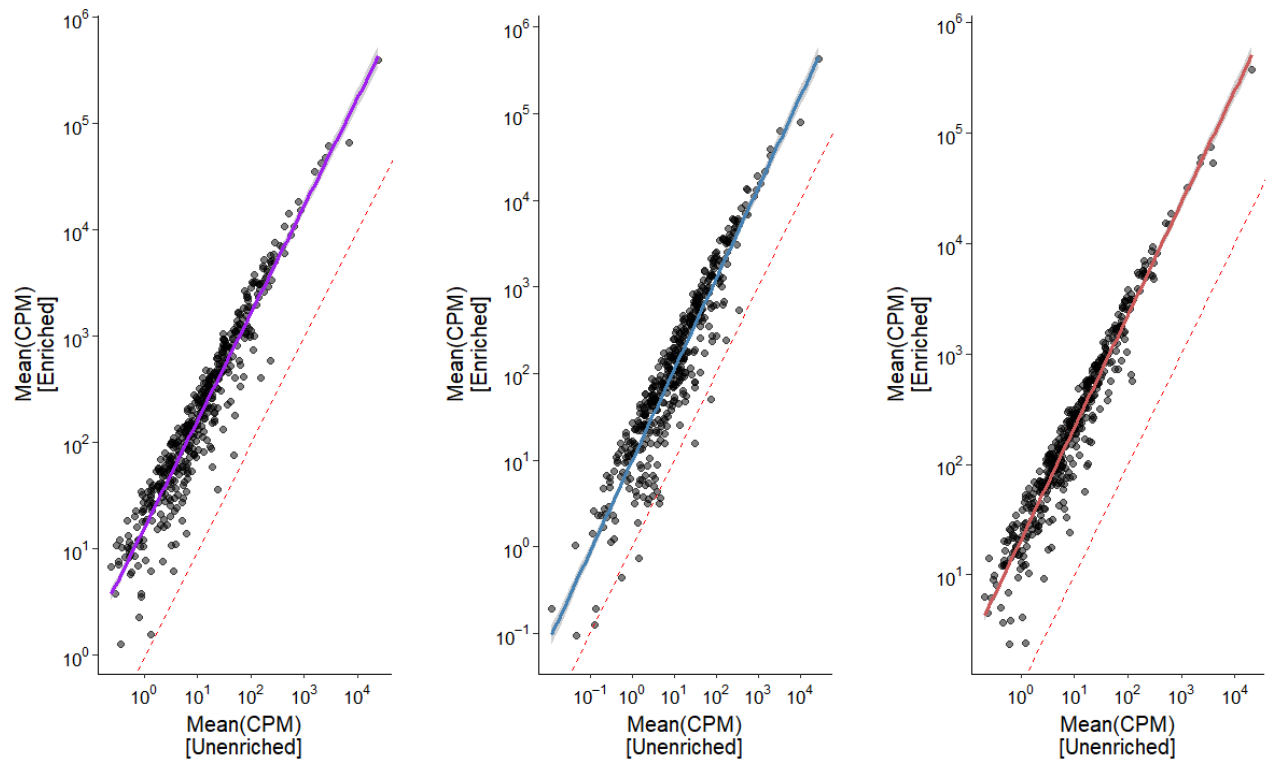
### Using Linear Regression and TOST to Determine Equal Proportional Representation

To evaluate whether the bait-capture enrichment distorted the proportional representation of the target genes (i.e., expression patterns were maintained), we fit linear regression models after plotting mean CPM for each transcript from the bait-capture enrichment data (henceforth “enriched”) against the conventional RNA-seq data (henceforth “unenriched”). Models were generated for the pooled dataset (total) and for the male and female subsets of the data independently (Figure 1). Before interpreting the model coefficients, we examined standard diagnostic plots to ensure the assumptions of linear regression were reasonably met. The residuals for all three models showed a slight departure from normality, but overall, the diagnostics indicated the assumptions were met.

Assuming proportionality is maintained in the enriched vs. unenriched data, the regression models should have slopes of approximately 1. In addition, because the bait-capture enrichment protocol significantly increases the counts for the target genes, at a much reduced sequencing depth, we expect to observe a clear positive shift in the intercepts. Consistent with these assumptions, all models have a positive y-intercept and slopes approximately parallel with the red-dashed line that represents a line with a slope of 1 and an intercept of 0 (Figure 1). The estimated intercepts and coefficients for each model are provided in Table 1.

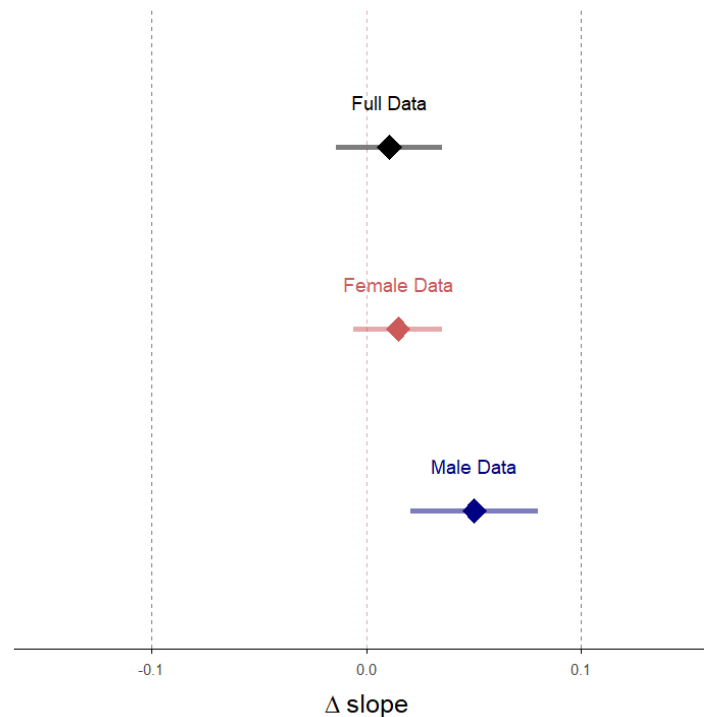
**Table 1: Linear model summaries and TOST results for mean CPM models.**

| Model       | Intercept | Intercept SE | Slope   | Slope SE | TOST Max                       | TOST Max                   | NHST                           |
|-------------|-----------|--------------|---------|----------|--------------------------------|----------------------------|--------------------------------|
|             |           |              |         |          | <i>p</i> -value<br>(Intercept) | <i>p</i> -value<br>(Slope) | <i>p</i> -value<br>(Intercept) |
| Full Data   | 1.19906   | 0.01953      | 1.0108  | 0.01491  | 1                              | <0.001                     | <0.001                         |
| Male Data   | 0.98942   | 0.02456      | 1.05023 | 0.01808  | 1                              | 0.003                      | <0.001                         |
| Female Data | 1.32267   | 0.01554      | 1.01474 | 0.01259  | 1                              | <0.001                     | <0.001                         |



**Figure 1. Linear regression models comparing bait-capture enriched and conventional RNA-seq counts show slopes near 1 and positive intercepts.** Mean CPM values for each target gene from enriched and unenriched samples are plotted for the pooled (purple), male (blue), and female (red) datasets, along with their corresponding fitted regression lines. Individual points represent each transcript, and the red dashed line represents a reference line with a slope of 1 and an intercept of 0.

To formally assess if the slopes are practically equivalent to one, we applied an equivalence test using the two one-sided t-test (TOST) framework. TOST allows users to define a range of values that they consider practically equivalent (or negligibly different) from the test value (Equation 1). Based on this range, t-tests are then used to test if the value is clearly less than, or greater than the defined equivalence bounds, which can be used as evidence of equivalence. Following the recommendations of Robinson et al., (2005), for regression-based equivalence regions, we defined 10% equivalence bounds to test if the slopes of each model were approximately equivalent to 1 (i.e., 0.9 to 1.1). For the pooled and female datasets, we observed that the difference between the estimated slope and 1 was not clearly different from zero, and they both fell within the 10% equivalence bounds region (Figure 2; Table 1). In contrast, the difference between the slope of the male dataset and 1, showed a small but statistically clear difference. However, because the difference still fell within the predefined 10% equivalence bounds, we concluded the effect was negligible. Using a simple one-sample t-test, we also observed that the intercepts were clearly greater than zero (Table 1).



**Figure 2. Slopes of the linear models are practically equivalent to 1, falling within the predefined 10% equivalence bounds.** A two one-sided t-test (TOST) was used to assess whether each model's slope was statistically equivalent to 1. Plotted are the differences between each estimated slope and 1, with 90% confidence intervals shown as error bars. Red dashed lines denote the positive and negative 10% equivalence bounds. Values whose confidence intervals lie entirely within these bounds are considered practically equivalent to 1.

Taken together, these results indicate that the slopes of the regression models are practically equivalent to 1, demonstrating that the proportionality of counts (i.e., the relative structure of the transcriptome) is generally maintained following bait-capture enrichment. The consistently positive intercepts further highlight a key advantage of the enrichment protocol. The higher mean CPM values in the enriched dataset are driven by both low sequencing depth and high transcript counts. Despite being sequenced at substantially lower depth, the enriched libraries still recovered a proportional representation of expression patterns, suggesting improved sensitivity for detecting true low-abundance transcripts. Because sequencing depth is directly tied to sequencing cost, this also underscores how bait-capture enrichment can reduce the cost per transcriptome when the research focus is restricted to a defined subset of genes (Vandesompele et al., 2002). These properties make bait-capture enrichment an attractive method for studying the chemical defenses of zebrafish.

### Using Relative Expression to Compare Individual Target Genes

Although proportionality appeared to be maintained at the global level, we reasoned that individual targets could still be disproportionately represented because bait-capture relies on a large number of unique probes. Directly comparing enriched and unenriched counts was not feasible due to substantial

differences in transcript abundances and sequencing depth between the two datasets. To address this, we normalized each transcript to a gene exhibiting stable counts within each dataset, allowing us to compare relative expression values across methods. This normalization enabled a gene-by-gene assessment of whether bait-capture enrichment distorted the representation of individual targets.

**Choosing a Stable Reference Gene using the Coefficient of Variation and geNorm Ranking**

A complication with this approach was that no conventional housekeeping transcripts were included in the target list, so we needed to identify a suitable reference from the dataset itself. A good reference must be stable, meaning it shows minimal between-sample variation. We first quantified stability using the coefficient of variation (CV), which measures how variable a transcript is relative to mean expression. Although selecting a reference gene solely from the subset with the lowest CV would likely be adequate for our purposes, we also applied the geNorm ranking algorithm, which is widely used for qRT-PCR reference gene validation (Yuan et al., 2025). Where CV captures the absolute variability, geNorm ranking evaluates pairwise expression variability. This refers to whether a gene maintains a consistent expression ratio to other candidates across samples. This ratio-based approach can identify genes that are proportionally co-regulated, which is another metric for a good reference gene. Therefore, after filtering by CV to remove noisy transcripts, we applied geNorm to assess the pairwise stability of the remaining candidates.

Using the combined dataset as an example, Table 2 shows the candidate list ranked by both CV and geNorm. Here, geNorm identified *TXN2* and *AKR1A1B* as the two most stable transcripts; however, *TXN2* ranked relatively poorly by CV, whereas *AKR1A1B* had the lowest combined CV rank. Because *AKR1A1B* was relatively stable across both metrics, we selected it as the reference gene for the full dataset. In practice, choosing among any of the twelve candidates would likely have little impact on downstream analyses, but using both CV and geNorm makes the choice more defensible and less arbitrary. Using the same process, *MGST1.2* and *SOD2* were chosen for looking at the male and female subsets, respectively (Supplementary Tables 1 & 2).

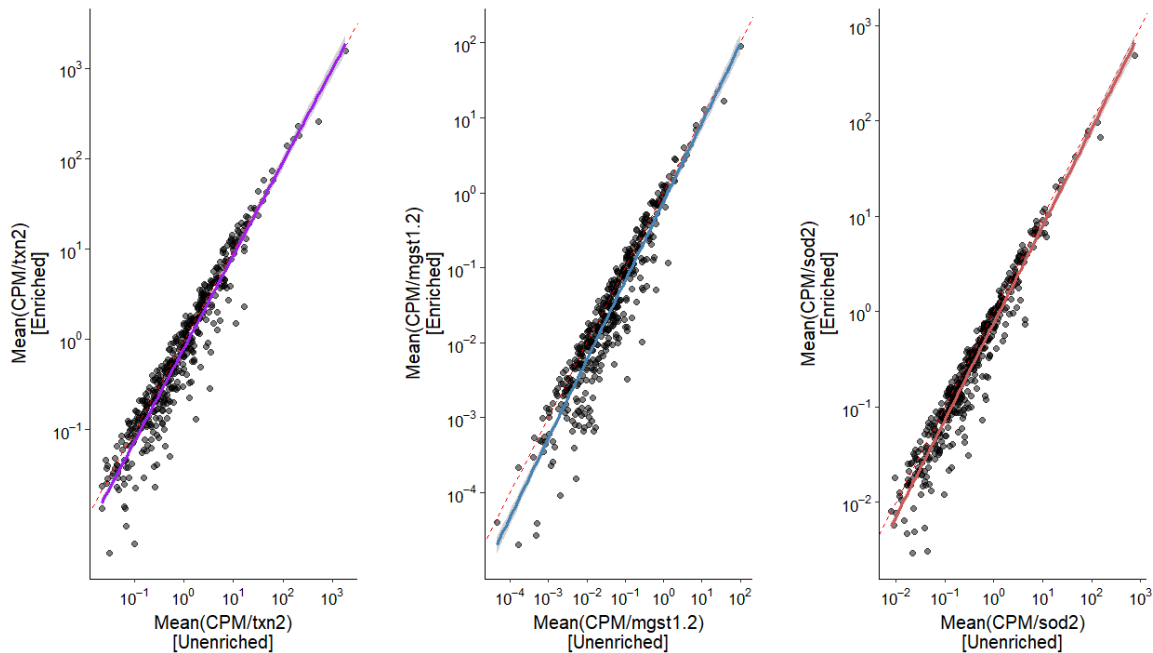
**Table 2: Using the Coefficient of Variation and geNorm ranking to choose a stable reference.**

| Gene_ID  | geNorm Rank<br>(Enriched) | geNorm Rank<br>(Unenriched) | geNorm Rank<br>(Combined) | CV Rank<br>(Enriched) | CV Rank<br>(Unenriched) | CV Rank<br>(Combined) |
|----------|---------------------------|-----------------------------|---------------------------|-----------------------|-------------------------|-----------------------|
| txn1     | 11                        | 11                          | 22                        | 11                    | 10                      | 21                    |
| clpp     | 10                        | 10                          | 20                        | 10                    | 11                      | 21                    |
| slc30a5  | 9                         | 5                           | 14                        | 12                    | 8                       | 20                    |
| abcg2a   | 8                         | 9                           | 17                        | 9                     | 5                       | 14                    |
| dnajb1a  | 7                         | 8                           | 15                        | 7                     | 1                       | 8                     |
| nat9     | 6                         | 6                           | 12                        | 8                     | 9                       | 17                    |
| slc22a15 | 5                         | 1                           | 6                         | 4                     | 3                       | 7                     |
| ugt2a3   | 4                         | 3                           | 7                         | 5                     | 4                       | 9                     |
| ephx5    | 3                         | 1                           | 4                         | 1                     | 6                       | 7                     |
| dnajc4   | 2                         | 7                           | 9                         | 3                     | 12                      | 15                    |
| akr1a1b  | 1                         | 4                           | 5                         | 2                     | 2                       | 4                     |
| txn2     | 1                         | 2                           | 3                         | 6                     | 7                       | 13                    |

### **Paired Equivalence Tests to Identify Transcripts with Equivalent Relative Expression**

After calculating relative expression values for the full, male, and female datasets using their respective stable reference genes (*AKR1A1B*, *MGST1.2*, *SOD2*), we fit linear models to compare the normalized enriched and unenriched RNA-seq data. Because both datasets were scaled to a relative value, we expected the intercepts to be practically equivalent to zero, and the slopes to be practically equivalent to 1. Once again, we checked diagnostic plots to ensure assumptions were reasonably met before interpreting the model coefficients.

As expected, across all three datasets, the slopes were practically equivalent to 1, within 10% equivalence bounds as described previously (Figure 3). In contrast, the intercepts for any model show a small but statistically clear negative deviation from zero, as determined by a one-tailed t-test ( $p < 0.001$ ; Table 3; Figure 3). Inferring from the regression models, this slight negative effect appears to be driven by transcripts with relatively low relative expression values, clustering below the regression line. This indicates that the relative expression values for the unenriched data are slightly larger than those for the enriched data, particularly for transcripts with low relative expression values. This suggests there may be slight differences in how low-abundance transcript counts are estimated in the enriched vs. unenriched data. It is possible this deviation arises from the bait-capture enrichment protocol providing better resolution of true low-abundance transcripts (Flint et al., 2023). In any case, using minimal-effects testing (MET), it was unclear if the negative intercept was outside the 10% equivalence bounds. Because the effect size was small and statistically inconclusive under the TOST and MET frameworks, we considered the deviation negligible and not of biological concern.



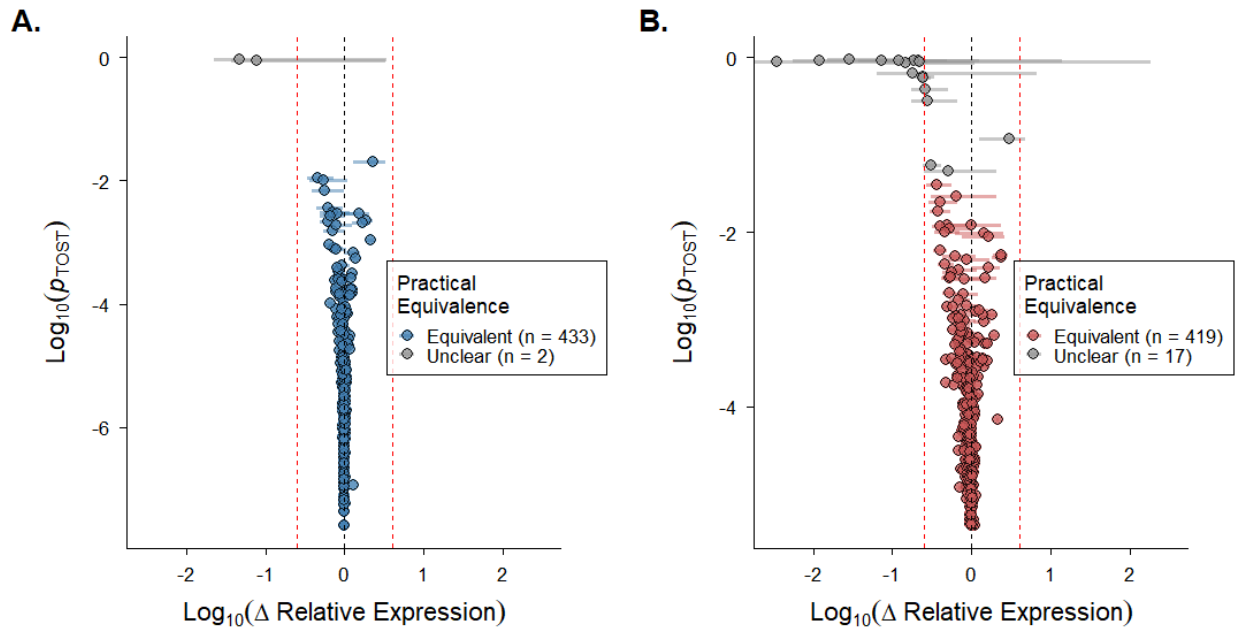
**Figure 3. Linear models derived from mean relative expression values show slopes near 1 and slightly negative intercepts.** Relative expression values for bait-capture enriched and conventional RNA-seq data were plotted for the pooled (purple), male (blue), and female (red) datasets, along with their corresponding fitted regression lines. Individual gene-level points are shown. The red dashed line

represents a reference line with a slope of 1 and an intercept of 0, included to illustrate proportionality between enriched and unenriched datasets after normalization.

**Table 3: TOST results for slopes and intercepts of relative expression linear models**

| Model       | Intercept | Intercept SE | Slope   | Slope SE | TOST Max                       | TOST Max                   |
|-------------|-----------|--------------|---------|----------|--------------------------------|----------------------------|
|             |           |              |         |          | <i>p</i> -value<br>(Intercept) | <i>p</i> -value<br>(Slope) |
| Full Data   | -0.10333  | 0.01173      | 1.03259 | 0.01492  | 0.612                          | < 0.001                    |
| Male Data   | -0.1201   | 0.03017      | 1.05072 | 0.01793  | 0.747                          | < 0.001                    |
| Female Data | -0.11976  | 0.01142      | 1.01865 | 0.01261  | 1                              | < 0.001                    |

We next performed paired TOSTs to assess the practical equivalence of individual transcripts between the enriched and unenriched data. Paired tests were appropriate because each enriched sample was derived from the same source material as its unenriched counterpart. For the equivalence bounds, we selected conservative limits of  $\pm 3$  units, reasoning that a difference of this magnitude would not be biologically meaningful when evaluating the effect of a hypothetical treatment on gene expression. Although this threshold is necessarily somewhat subjective, we consulted with our supervisors, who agreed that  $\pm 3$  units were reasonable boundaries. Because  $\sim 435$  equivalence tests were performed, we controlled the false discovery rate (FDR) at 5% using the Benjamini–Hochberg procedure. We also constructed FDR-adjusted confidence intervals using the method described by Benjamini and Yekutieli (2005).



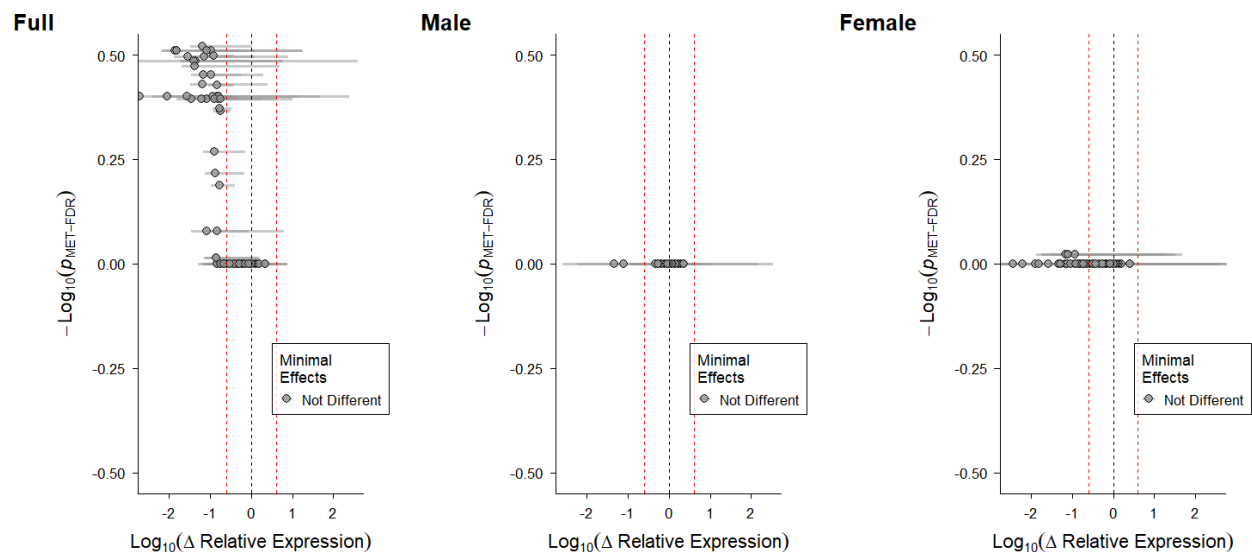
**Figure 4. Most transcripts show practically equivalent relative expression between bait-capture enriched and unenriched RNA-seq datasets.** Volcano plots display paired TOST results for the male (A) and female (B) datasets. Each transcript is plotted by  $\text{log}_{10}(p_{\text{TOST-Max}})$  on the y-axis and  $\text{log}_{10}(\text{RE}_{\text{enriched}} - \text{RE}_{\text{unenriched}})$  on the x-axis.

$RE_{\text{unenriched}}$ ) on the x-axis, with points coloured according to their equivalence classification. Horizontal bars represent FDR-adjusted 90% confidence intervals (FDR = 5%). Red dashed lines denote the positive and negative equivalence bounds, and the black dashed line marks zero difference in relative expression.

After performing the paired TOSTs and adjusting for multiple comparisons, we found that the vast majority of transcripts were clearly equivalent with respect to their relative expression values (Figure 4AB). In the combined dataset, only 75 transcripts were not clearly equivalent within the  $\pm 3$ -unit bounds, despite the presence of sex-specific expression differences for many genes, which broadened the 90% confidence intervals (Supplementary Figure 1). When the male and female datasets were analyzed separately, only 2 transcripts in the male dataset and 17 transcripts in the female dataset failed to be clearly within the defined equivalence bounds (Figure 4AB). These results indicate that, at the individual-gene level, bait-capture enrichment preserves relative expression for nearly all targets.

### **Minimal Effects Tests to Determine if Transcripts are Clearly ‘Not Equivalent’**

Because failing to reject the null hypothesis in a TOST does not provide evidence of non-equivalence, we next performed minimal-effects tests (METs). METs are conceptually similar to TOSTs, but instead of testing whether a difference lies within a predefined equivalence range, they test whether a difference is clearly outside that range. This allowed us to evaluate whether any transcript showed a relative expression difference between the enriched and unenriched datasets that exceeded our  $\pm 3$ -unit equivalence bounds. After adjusting p-values for multiple testing using the Benjamini–Hochberg procedure, no transcripts were found to be clearly outside the equivalence bounds (Figure 5). These results indicate that although a small number of transcripts were not classified as equivalent by TOST, none showed statistically supported evidence of non-equivalence.



**Figure 5. No transcripts show clear evidence of relative expression differences outside the predefined equivalence bounds.** Volcano plots display minimal-effects test (MET) results for the full, male, and female datasets. Each transcript was tested for a clear departure from the  $\pm 3$ -unit equivalence bounds used in the TOST analyses. BH-adjusted p-values are plotted against the difference in relative

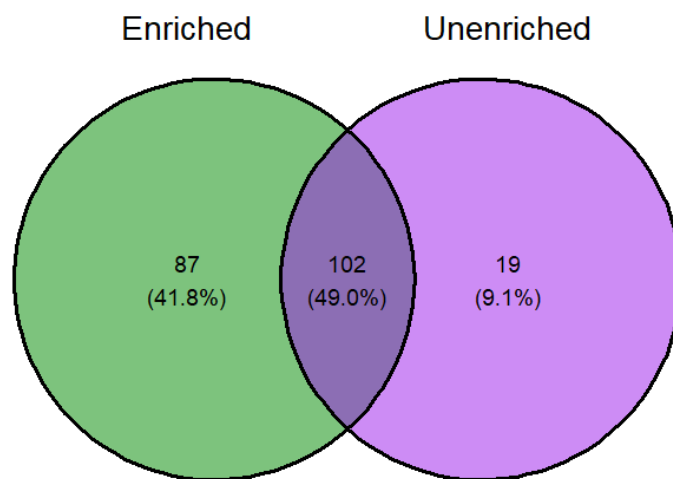


expression between enriched and unenriched datasets, with horizontal bars representing FDR-adjusted 90% confidence intervals (FDR = 5%). Red dashed lines denote the positive and negative equivalence bounds, and the black dashed line marks zero difference. No transcripts in any dataset were found clearly outside the equivalence bounds.

### **Differential Gene Expression Analyses of Male and Female Zebrafish Livers using Enriched and Unenriched RNA-Sequencing Data**

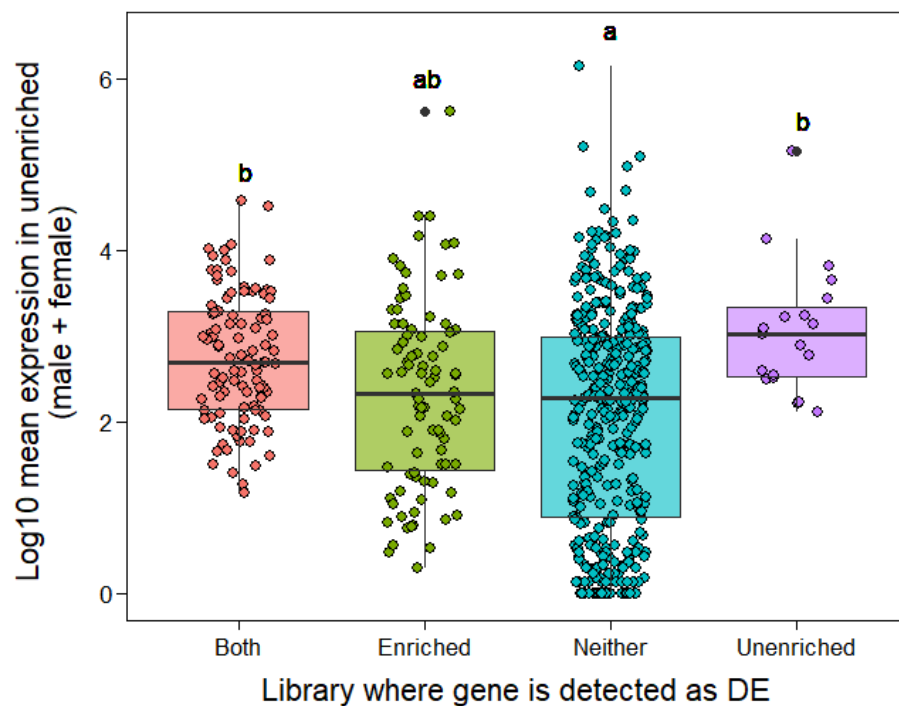
Because we observed no significant distortion of the bait-capture enrichment data in our regression-based equivalence test for proportionality or transcript-level equivalence testing using relative expression values, we reasoned that differential expression analyses of enriched and unenriched datasets may yield similar results. However, as bait-capture enrichment supposedly provides greater resolution, particularly for low-abundance transcripts, we anticipated that differential expression analysis using the bait-enriched data may reveal additional differentially expressed genes (DEGs) not detected in the unenriched dataset (Flint et al., 2023). Therefore, our goal was to compare DEGs identified in each dataset and determine whether bait-capture enrichment provides increased sensitivity or clarity, particularly among low-abundance transcripts.

To address this goal, we used the DESeq2 package to identify DEGs between male and female zebrafish in both the enriched and unenriched libraries. Using a fold-change cutoff of 2 and filtering for p-values < 0.05, we identified 189 DEGs in the enriched dataset and 121 DEGs in the unenriched dataset (Figure 6). Of these, 102 DEGs (49%) were identified in both libraries, 87 (41.8%) were unique to the enriched dataset, and 19 (9.1%) were unique to the unenriched dataset (Figure 6). Considering that both datasets are generated from the same source tissues, these results indicate that bait-capture enrichment is capable of detecting more DEGs than conventional RNA-sequencing.



**Figure 6. Enriched libraries show more differentially expressed genes than unenriched libraries.** Venn diagram shows the number of differentially expressed genes between male and female zebrafish livers identified using enriched and unenriched sequencing libraries. The number of differentially expressed genes is shown as a number and percentage of the total.

Because we have shown that bait-capture enrichment does not appear to significantly distort the data, we considered whether the increased number of DEGs in the enriched data was driven by better resolution of low-abundance transcripts. To examine this, we sorted DEGs based on whether they were differentially expressed in either dataset, both datasets, or neither, and compared the mean counts for all DEGs using the unenriched data. We observed that DEGs found uniquely in the unenriched dataset had greater mean counts than DEGs found uniquely in the enriched dataset. Additionally, mean counts for DEGs identified uniquely in the enriched dataset were lower than those of DEGs identified in both. Although these differences were statistically unclear, DEGs identified uniquely in the enriched dataset clearly included many transcripts with lower mean counts than observed in the unenriched dataset (i.e., lower range). This suggests that bait-capture enrichment may be better suited for identifying DEGs with truly low-abundance transcripts.



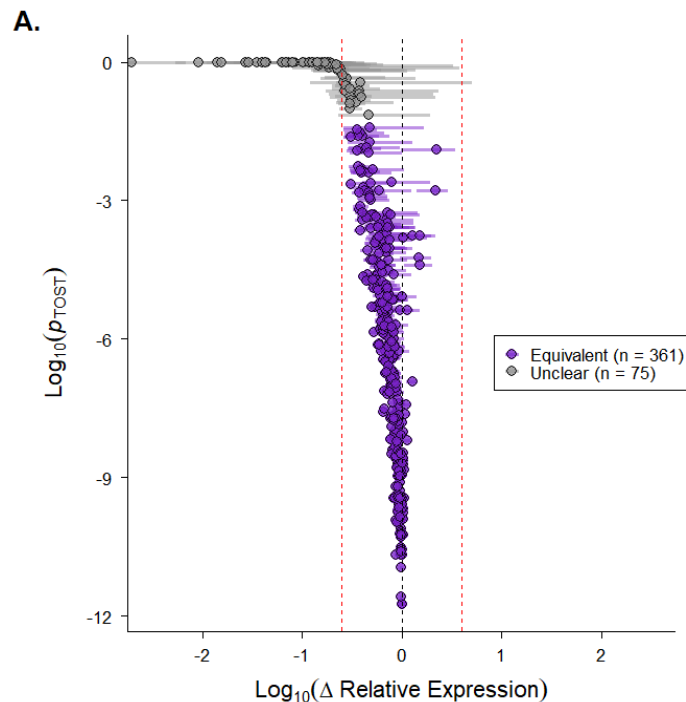
**Figure 7. Genes detected as differentially expressed in enriched libraries have lower average read counts.** Box plots displays the average  $\log_{10}$  read counts of genes from the unenriched library. Each transcript is plotted on the y-axis by the average read counts across male and female samples and is categorized on the x-axis by which library it was deemed to be differentially expressed between males and females. Different letters indicate statistically clear differences as determined by one-way ANOVA and Tukey's HSD ( $n = 102, 87, 387, 19$ ).

## Conclusion

Taken together, our results demonstrate that the bait-capture enrichment protocol produces comparable transcriptomic data to conventional RNA-seq. Across all analyses, we detected practically equivalent proportional representation of counts, and no transcripts that were clearly distorted in the enriched dataset

(Figures 4, 5). Moreover, the enrichment protocol may provide improved resolution for low-abundance transcripts and increased sensitivity in differential expression analyses (Figure 7). These findings indicate that bait-capture enrichment is not only defensible to use but may offer clear advantages in the context of the chemical defensome. We therefore recommend the use of this protocol and the associated bait set for future studies.

## Supplementary Information



**Supplementary Figure 1: Most transcripts show practically equivalent relative expression between bait-capture enriched and unenriched RNA-seq datasets.** Volcano plots display paired TOST results for the full dataset. Each transcript is plotted by  $\log_{10}(p_{\text{TOST-Max}})$  on the y-axis and  $\log_{10}(\text{RE}_{\text{enriched}} - \text{RE}_{\text{unenriched}})$  on the x-axis, with points coloured according to their equivalence classification. Horizontal bars represent FDR-adjusted 90% confidence intervals (FDR = 5%). Red dashed lines denote the positive and negative equivalence bounds, and the black dashed line marks zero difference in relative expression.

**Supplementary Table 1: Male dataset geNorm and CV Ranking**

| Gene_ID        | geNorm Rank<br>(Enriched) | geNorm Rank<br>(Unenriched) | geNorm Rank<br>(Combined) | CV Rank<br>(Enriched) | CV Rank<br>(Unenriched) | CV Rank<br>(Combined) |
|----------------|---------------------------|-----------------------------|---------------------------|-----------------------|-------------------------|-----------------------|
| cyp4v2b        | 9                         | 2                           | 11                        | 8                     | 3                       | 11                    |
| txna           | 8                         | 1                           | 9                         | 7                     | 2                       | 9                     |
| abce1          | 7                         | 8                           | 15                        | 9                     | 7                       | 16                    |
| slc31a2        | 6                         | 9                           | 15                        | 5                     | 5                       | 10                    |
| hspa8          | 5                         | 7                           | 12                        | 6                     | 9                       | 15                    |
| adh8a          | 4                         | 6                           | 10                        | 4                     | 4                       | 8                     |
| adh5           | 3                         | 3                           | 6                         | 1                     | 1                       | 2                     |
| cyp4t8         | 2                         | 5                           | 7                         | 2                     | 8                       | 10                    |
| <b>mgst1.2</b> | 1                         | 4                           | <b>5</b>                  | 3                     | 6                       | <b>9</b>              |

**Supplementary Table 2: Female dataset geNorm and CV Ranking**

| Gene_ID     | geNorm Rank<br>(Enriched) | geNorm Rank<br>(Unenriched) | geNorm Rank<br>(Combined) | CV Rank<br>(Enriched) | CV Rank<br>(Unenriched) | CV Rank<br>(Combined) |
|-------------|---------------------------|-----------------------------|---------------------------|-----------------------|-------------------------|-----------------------|
| dnajc19     | 12                        | 2                           | 14                        | 5                     | 7                       | 12                    |
| aldh2.2     | 11                        | 9                           | 20                        | 8                     | 5                       | 13                    |
| abcc12      | 10                        | 5                           | 15                        | 12                    | 4                       | 16                    |
| naa60       | 9                         | 11                          | 20                        | 4                     | 9                       | 13                    |
| aldh2.1     | 8                         | 10                          | 18                        | 2                     | 6                       | 8                     |
| hspa14      | 7                         | 3                           | 10                        | 3                     | 10                      | 13                    |
| rxrbb       | 6                         | 7                           | 13                        | 1                     | 2                       | 3                     |
| keap1b      | 5                         | 6                           | 11                        | 6                     | 3                       | 9                     |
| <b>sod2</b> | 4                         | 1                           | <b>5</b>                  | 10                    | 8                       | <b>18</b>             |
| cyp4t8      | 3                         | 4                           | 7                         | 11                    | 1                       | 12                    |
| txn1        | 2                         | 8                           | 10                        | 7                     | 11                      | 18                    |
| fthl31      | 1                         | 12                          | 13                        | 9                     | 12                      | 21                    |

**Supplementary Table 3: Tukey's HSD results for comparison of mean CPMs**

|                            | Log10 Difference | Lower Bound  | Upper Bound  | Adjusted p-value |
|----------------------------|------------------|--------------|--------------|------------------|
| <b>Enriched-Both</b>       | -0.430011461     | -0.869437087 | 0.009414165  | 0.057717         |
| <b>Neither-Both</b>        | -0.68177509      | -1.017548132 | -0.346002048 | 0.000001         |
| <b>Unenriched-Both</b>     | 0.324632623      | -0.427734811 | 1.077000057  | 0.682473         |
| <b>Neither-Enriched</b>    | -0.251763629     | -0.609630261 | 0.106103003  | 0.268382         |
| <b>Unenriched-Enriched</b> | 0.754644084      | -0.007839848 | 1.517128016  | 0.053588         |
| <b>Unenriched-Neither</b>  | 1.006407713      | 0.298572703  | 1.714242722  | 0.001545         |

## References

- Eide, M., Zhang, X., Karlsen, O. A., Goldstone, J. V., Stegeman, J., Jonassen, I., & Goksøyr, A. (2021). The chemical defensome of five model teleost fish. *Scientific Reports*, 11(1), 10546. <https://doi.org/10.1038/s41598-021-89948-0>
- Flint, A., Cooper, A., Rao, M., Weedmark, K., Carrillo, C., & Tamber, S. (2023). Targeted metagenomics using bait-capture to detect antibiotic resistance genes in retail meat and seafood. *Frontiers in Microbiology*, 14. <https://doi.org/10.3389/fmicb.2023.1188872>
- Goldstone, J. V. (2008). Environmental sensing and response genes in Cnidaria: The chemical defensome in the sea anemone *Nematostella vectensis*. *Cell Biology and Toxicology*, 24(6), 483–502. <https://doi.org/10.1007/s10565-008-9107-5>
- Goldstone, J. V., Hamdoun, A., Cole, B. J., Howard-Ashby, M., Nebert, D. W., Scally, M., Dean, M., Epel, D., Hahn, M. E., & Stegeman, J. J. (2006). The chemical defensome: Environmental sensing and response genes in the *Strongylocentrotus purpuratus* genome. *Developmental Biology*, 300(1), 366–384. <https://doi.org/10.1016/j.ydbio.2006.08.066>
- Kime, D. E. (1999). A strategy for assessing the effects of xenobiotics on fish reproduction. *The Science of the Total Environment*, 225(1–2), 3–11. [https://doi.org/10.1016/s0048-9697\(98\)00328-3](https://doi.org/10.1016/s0048-9697(98)00328-3)
- Piowowska, D., & Kiedrzyńska, E. (2022). Xenobiotics as a contemporary threat to surface waters. *Ecology & Hydrobiology*, 22(2), 337–354. <https://doi.org/10.1016/j.ecohyd.2021.09.003>
- Robinson, A. P., Duursma, R. A., & Marshall, J. D. (2005). A regression-based equivalence test for model validation: Shifting the burden of proof. *Tree Physiology*, 25(7), 903–913. <https://doi.org/10.1093/treephys/25.7.903>
- Sakkos, A., Saint-John, B., Tymł, T., Myskova, E., Aureli, L., Inman, J. L., Snijders, A. M., Mouncey, N. J., Mukundan, H., & Schulz, F. (2026). Agnostic capture of pathogens for the detection and diagnostics of emerging threats. *iScience*, 29(2), 114684. <https://doi.org/10.1016/j.isci.2026.114684>

Vandesompele, J., De Preter, K., Pattyn, F., Poppe, B., Van Roy, N., De Paepe, A., & Speleman, F. (2002).

Accurate normalization of real-time quantitative RT-PCR data by geometric averaging of multiple internal control genes. *Genome Biology*, 3(7), research0034.1-research0034.11.

<https://doi.org/10.1186/gb-2002-3-7-research0034>

Yuan, Y., Liu, C., Bao, J., & Li, F. (2025). Selection and Validation of Appropriate Reference Genes for

qRT-PCR Analysis of *Iris germanica* L. Under Various Abiotic Stresses. *Food Science &*

*Nutrition*, 13(1), e4765. <https://doi.org/10.1002/fsn3.4765>